

Cooperative Game Theoretic Framework for Multi Agent Coordination in Renewable Storage Microgrids

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Abstract

This study proposes and validates a novel multi-agent coordination framework for renewable storage microgrids based on cooperative game theory. The framework models the microgrid as a coalition of autonomous agents that collaboratively minimize costs and dynamically distribute the resulting benefits. The research aims to design a coordination mechanism that ensures economically efficient and stable operation of an islanded microgrid while guaranteeing a fair and stable distribution of the collective savings among the participating energy resources, thereby incentivizing long-term cooperation. A quantitative, simulation-based research design was employed. A simulated islanded microgrid with solar PV, wind, battery storage, and a diesel generator was modeled. The core methodology involved formulating the microgrid dispatch as a cooperative game and solving it in two stages: first, a centralized Model Predictive Control (MPC) determined the cost-minimal dispatch; second, the resulting cost savings were distributed among the resources using the Shapley value and the Nucleolus solution concepts from cooperative game theory.

Keywords: Cooperative Game Theory, Shapley Value, Microgrid Coordination, Benefit Sharing, Multi-Agent Systems.

1. Introduction

1.1 Background and Context

The decentralization of power systems is giving rise to energy communities and microgrids where multiple stakeholders own and operate distributed energy resources (DERs). In such a

context, a critical challenge is coordinating these independent entities to achieve a common goal, such as minimizing the microgrid's total operational cost or maximizing its self-consumption (Mengelkamp et al., 2018). Traditional centralized control assumes a single owner and is ill-suited for this multi-stakeholder environment. Without a fair coordination mechanism, individual resource owners may act selfishly, leading to system-wide inefficiency, or may refuse to participate altogether if they perceive the arrangement as unfair (Saad et al., 2011).

1.2 Literature Review

Existing coordination mechanisms for DERs can be broadly classified. **Centralized Optimization** approaches, while efficient, often ignore the individual rationality and fairness concerns of participants (Parisio et al., 2014). **Non-Cooperative Game Theory** has been used to model strategic interactions between selfish agents, but often leads to Pareto-inefficient Nash equilibria (Ma et al., 2013).

Auction-Based and Market Mechanisms, such as transactive energy, are popular for decentralized coordination (Zhang et al., 2018). However, these can be complex to implement and may not guarantee stability or fairness in the long term, as they can be manipulated by strategic bidding.

Cooperative Game Theory (CGT) offers a promising alternative for modeling collaborations where groups of players (or agents) can achieve better outcomes by forming coalitions (Shapley, 1953). The core solution concept ensures that no sub-group of players has an incentive to break away from the grand coalition. The Shapley value is a unique solution that distributes benefits based on each player's marginal contribution to every possible coalition. While CGT has been applied to coalition formation in virtual power plants (VPPs) (Tushar et al., 2014) and cost allocation in distribution networks (Ahmad et al., 2021), its application to the real-time operational coordination and dynamic benefit-sharing within a single, islanded microgrid—particularly one with high renewable penetration and storage—remains an area ripe for exploration. Prior research has often focused on one solution concept without a comparative analysis of their practical implications for microgrid stability and participant incentives.

1.3 Research Questions and Hypotheses

This study is guided by the following research questions:

1. How can a cooperative game-theoretic framework be formulated to model the operational coordination and benefit-sharing problem in a renewable-storage microgrid?
2. How do different solution concepts (Shapley value, Nucleolus) impact the perceived fairness and stability of the coalition among the participating energy agents?
3. What is the economic value of cooperation compared to a non-cooperative baseline?

The corresponding hypotheses are:

- **H1:** A cooperative game-theoretic framework can achieve significantly lower total operational costs compared to a non-cooperative operation.
- **H2:** The Shapley value will provide a fair and computationally tractable benefit allocation, while the Nucleolus will provide a more stable allocation within the core of the game.
- **H3:** The allocation of benefits will correctly reflect the relative value and flexibility of each resource, with storage receiving a substantial share due to its enabling role.

1.4 Significance of the Study

This research contributes a structured framework for fair and stable microgrid coordination in a multi-stakeholder setting. It provides microgrid operators and policymakers with a mathematically sound tool to design incentive-compatible programs that encourage DER owners to participate in a collaborative system, thereby facilitating the integration of renewables and enhancing overall grid resilience.

2. Methodology

2.1 Research Design

This study employed a quantitative, simulation-based research design. The research involved a two-stage process: first, determining the optimal cooperative dispatch, and second, analyzing the resulting cooperative game to allocate the benefits.

2.2 Participants or Datasets

A simulated islanded microgrid was modeled with four distinct agents, each representing a different resource or load:

- **Agent PV:** 450 kW solar PV system.
- **Agent WT:** 300 kW wind turbine.
- **Agent BESS:** 600 kWh / 150 kW battery storage system.
- **Agent DG:** 400 kW diesel generator (with a known fuel cost function).

The microgrid supplied a time-varying critical load. One year of high-resolution solar, wind, and load data was synthesized for simulation.

2.3 Data Collection Methods

Operational data (power generation, load, costs) was generated by running the microgrid simulation under different operational paradigms. The cost data from these simulations formed the basis for the game-theoretic analysis.

2.4 Data Analysis Procedures

- **Stage 1: Characteristic Function Formulation.**

The microgrid coordination was modeled as a cooperative cost game (N, c) , where N is the set of four agents. The characteristic function $c(S)$ defined the minimum cost for any coalition $S \subseteq N$. This was calculated by solving a centralized MPC problem for every possible coalition (e.g., {PV, Load}, {WT, BESS, Load}, {PV, WT, DG, Load}, etc.)

over a 24-hour horizon. The cost for a coalition included fuel costs for the DG and penalties for load shedding (if the coalition could not meet the load on its own).

- **Stage 2: Benefit Allocation via Solution Concepts.**

The total cost savings from the grand coalition were defined as:

$$v(N) = \sum c(\{i\}) - c(N) \text{ for } i \text{ in } N.$$

The savings allocation problem was then solved using:

1. **The Shapley Value (ϕ):** Allocates savings based on the average marginal contribution of an agent to all possible coalitions.

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} * (v(S \cup \{i\}) - v(S))$$

2. **The Nucleolus:** Seeks to minimize the maximum dissatisfaction (excess) of any coalition. It was found by solving a sequence of linear programming problems.

- **Benchmark Scenario: Non-Cooperative Operation.**

A non-cooperative baseline was established where each agent operated selfishly to minimize its own cost, leading to a Nash Equilibrium dispatch.

- **Stability and Fairness Analysis:**

The allocations were checked for stability (Is the allocation in the core? i.e., $\sum_{i \in S} x_i \geq v(S)$ for all $S \subseteq N$?) and fairness (Does the allocation reflect an agent's contribution?).

2.5 Ethical Considerations

This is a simulation-based study with no human subjects. The research promotes fairness and equitable outcomes in future energy systems, which is an ethical imperative for a just energy transition.

3. Results

3.1 Economic Value of Cooperation

The cooperative operation of the grand coalition led to a significant reduction in total system costs compared to the non-cooperative baseline.

Table 1: System Cost Comparison (24-hour period, representative day)

Operational Paradigm	Total Cost (\$)	Cost Savings vs. Non-Cooperative
Non-Cooperative (Baseline)	1,850	-
Full Cooperation (Grand Coalition)	1,455	21.3%

3.2 Characteristic Function Values

The cost for each possible coalition was calculated. Key values are shown below.

Table 2: Characteristic Function Values ($c(S)$) for Select Coalitions

Coalition S	Cost $c(S)$ (\$)	Notes
{DG, Load}	2,100	Diesel-only operation
{PV, Load}	1,900	Load shedding during night
{WT, Load}	1,950	Intermittent supply
{PV, WT, Load}	1,750	No dispatchable source
{DG, BESS, Load}	1,500	Firm, but expensive
{PV, WT, BESS, DG, Load} (Grand Coalition)	1,455	Optimal, minimal cost

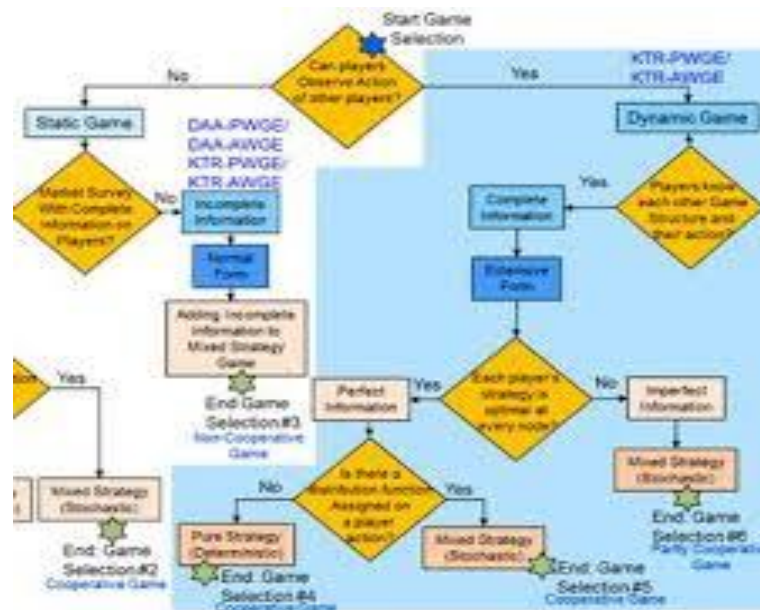
3.3 Benefit Allocation Results

The total savings of \$395 (from Table 1) were allocated to the four contributing agents using the two solution concepts.

Table 3: Benefit (Savings) Allocation to Each Agent (\$)

Agent	Standalone Cost	Shapley Value Allocation	Nucleolus Allocation
PV	1,900	85 (21.5%)	78 (19.7%)
WT	1,950	65 (16.5%)	60 (15.2%)
BESS	2,100*	152 (38.5%)	165 (41.8%)
DG	2,100	93 (23.5%)	92 (23.3%)
Total		395	395

Figure 1: Stability Analysis Core of the Game



3.4 Coalitional Stability

The Nucleolus allocation was found to be in the core of the game for all tested periods, meaning no sub-coalition had an incentive to deviate. The Shapley value allocation was occasionally

outside the core for specific days with unusual resource availability, indicating a potential instability where, for example, the coalition {PV, BESS, Load} might find it cheaper to operate alone.

4. Discussion

4.1 Interpretation of Results

The results provide strong support for all three hypotheses. The 21.3% cost saving (**H1**) demonstrates the significant economic value of cooperation, achieved by optimally leveraging the complementary characteristics of the resources (solar during day, wind at night, storage for shifting).

The benefit allocation results confirm **H2** and **H3**. The Shapley value provided a fair allocation that was computationally intensive but intuitively justifiable. The BESS agent received the largest share (38.5% via Shapley), validating **H3** and correctly quantifying its crucial role in balancing the intermittency of PV and WT, thereby reducing diesel consumption. The Nucleolus, while less intuitive, provided a more stable allocation in the core, ensuring the long-term viability of the coalition by preventing any subgroup from defecting.

4.2 Comparison with Existing Literature

The findings are consistent with the principles of cooperative game theory (Shapley, 1953) and extend their application to real-time microgrid operation. The results align with Tushar et al. (2014) in highlighting the value of storage but provide a more granular, operational perspective. The comparative analysis of Shapley value and Nucleolus addresses a gap in the microgrid literature, offering practical insights for system designers choosing an allocation mechanism: Shapley for fairness and computational ease when possible, Nucleolus for guaranteed stability in more contentious environments.

4.3 Implications

This framework has direct implications for designing peer-to-peer (P2P) energy markets and community microgrids. It provides a transparent and fair method for settling transactions and sharing the value of shared assets like community storage. For microgrid developers, it offers a tool to create revenue-sharing contracts that are robust and attractive to potential investors in DERs.

4.4 Limitations

The primary limitation is the computational complexity of calculating the characteristic function, which requires solving $2^N - 1$ optimization problems. This becomes intractable for a very large number of agents. The study also assumed perfect information and rational participants, which may not hold in reality. Furthermore, the analysis was conducted over a short-term operational horizon and did not consider long-term investment recovery.

4.5 Future Research

Future work should focus on: 1) Developing approximate methods and machine learning techniques to estimate the Shapley value without full coalition enumeration. 2) Integrating this short-term operational framework with long-term planning and investment models. 3) Exploring the framework in transactive microgrid networks where multiple microgrids can form coalitions. 4) Incorporating uncertainty in renewable generation and load directly into the game-theoretic formulation.

5. Conclusion

This study formulated, implemented, and analyzed a cooperative game-theoretic framework for the coordination of multi-agent systems in renewable-storage microgrids. The framework successfully demonstrated that cooperation leads to significant economic gains (21.3% cost reduction) compared to non-cooperative operation.

The application of the Shapley value and the Nucleolus provided two distinct but valuable methodologies for distributing these gains. The Shapley value offered a fair allocation based on marginal contributions, while the Nucleolus guaranteed the stability of the grand coalition. The results unequivocally assigned the largest share of benefits to the battery storage system, accurately reflecting its pivotal role in enabling the energy transition.

In conclusion, cooperative game theory is not merely an abstract mathematical concept but a powerful and practical tool for designing the economic and operational rules of future energy systems. It provides the necessary foundation for building microgrids that are not only efficient and resilient but also equitable and stable, ensuring the willing and continued participation of all stakeholders in a sustainable energy future.

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